

ENGINEERING PORTFOLIO · 2026

Muhammet Mert Erdoğan

Mechanical engineer working on design, structural analysis and manufacturing. BSc Mechanical Engineering, thesis MSc in progress at Istanbul University-Cerrahpaşa. Certified SOLIDWORKS Professional (CSWP).

mmert1erdogan@gmail.com linkedin.com/in/mmerterdogan mmerterdogan.github.io

Shell theory said 11.2. The real head said 3.05.

Mechanical System Design project, İÜC · Oct 2025 - Jan 2026
Advisor: Dr. Murat Bozkurt · individual work

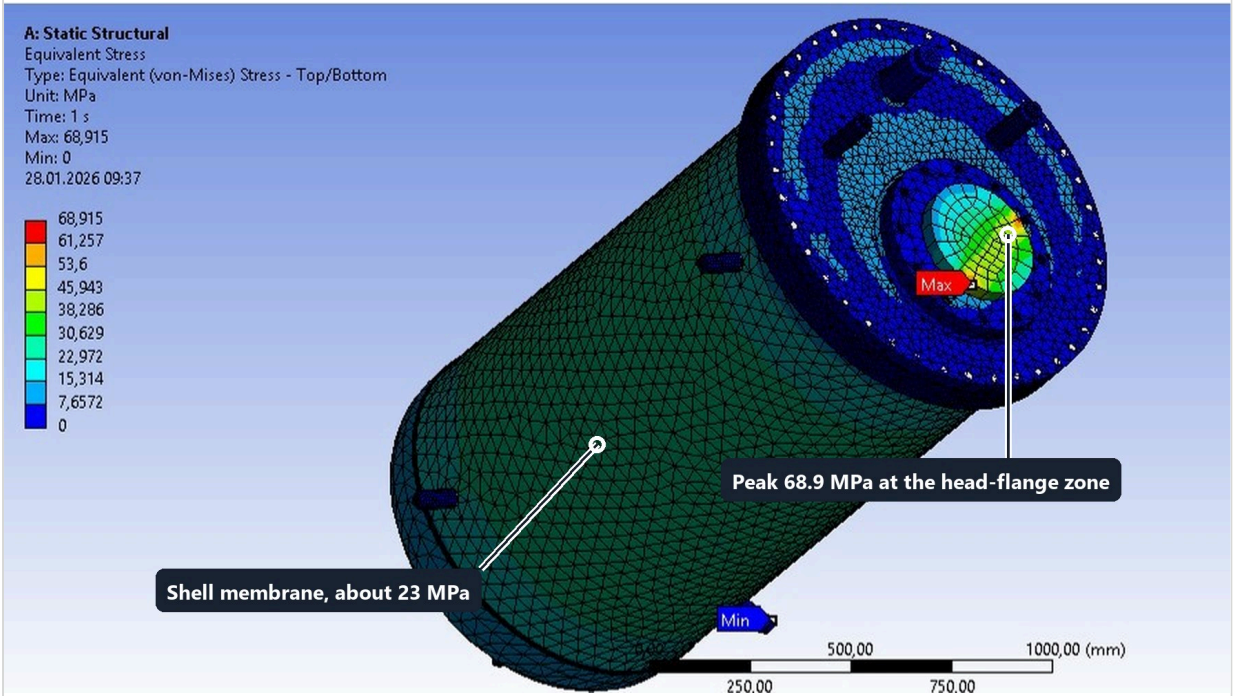
I calculated the stresses of a pressure vessel by hand, built an ANSYS model of the vessel and its bolted head and showed that the head-flange region, not the shell, sets the safety margin.

3.05
minimum safety factor (FEA, at the head)

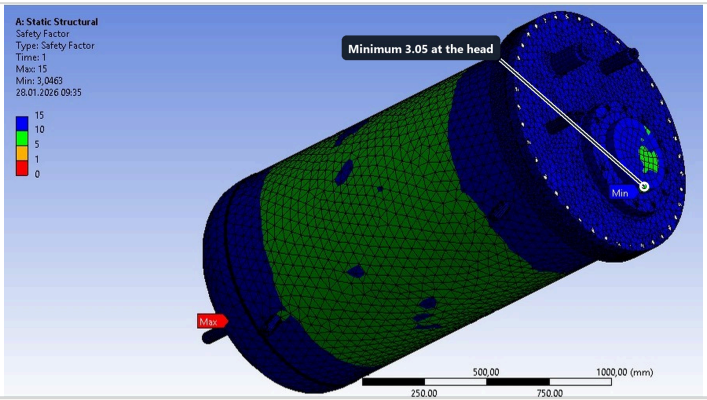
11.2
safety factor of the shell membrane (theory)

68.9 MPa
peak von Mises stress

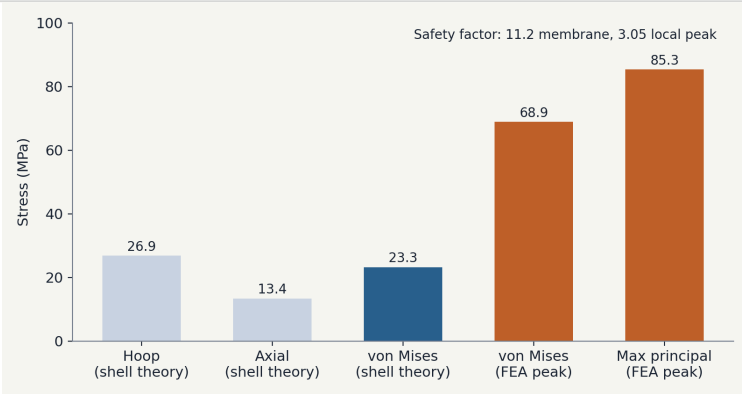
45.8 m
worst-case blast safety distance



Equivalent stress, ANSYS Mechanical 2022 R1



Safety factor map



Shell theory against FEA peak values

Two routes, compared at the right level

1 · HAND CALCULATION
Hoop, axial and von Mises stress, radial growth

→

2 · FEA MODEL
Vessel and bolted head, linear static, refined mesh

→

3 · TWO-LEVEL CHECK
Membrane vs theory, peaks at discontinuities

→

4 · CONSEQUENCE
Stored energy, TNT equivalent, safety distance

PROBLEM

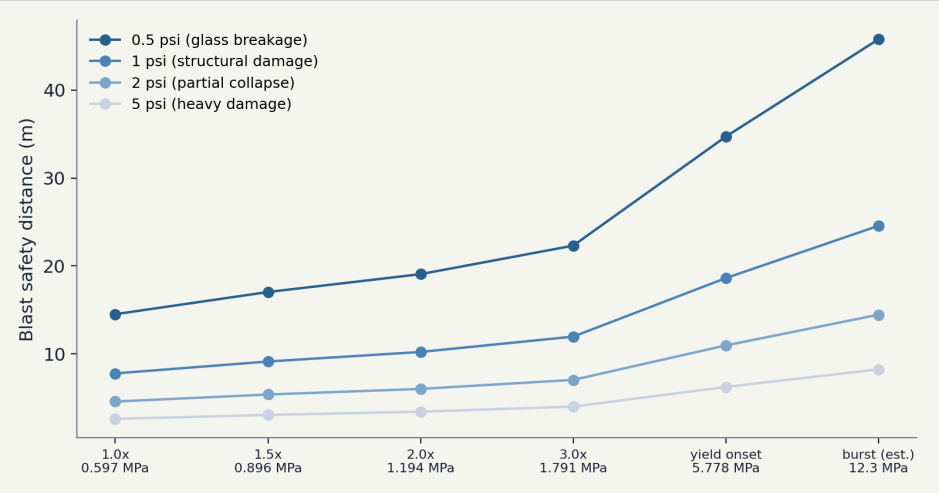
Shell r = 450 mm, t = 10 mm, head 12 mm, L = 1600 mm, SA-516 Gr.70 (yield 260 MPa), internal pressure 0.597 MPa.

WHAT I DID

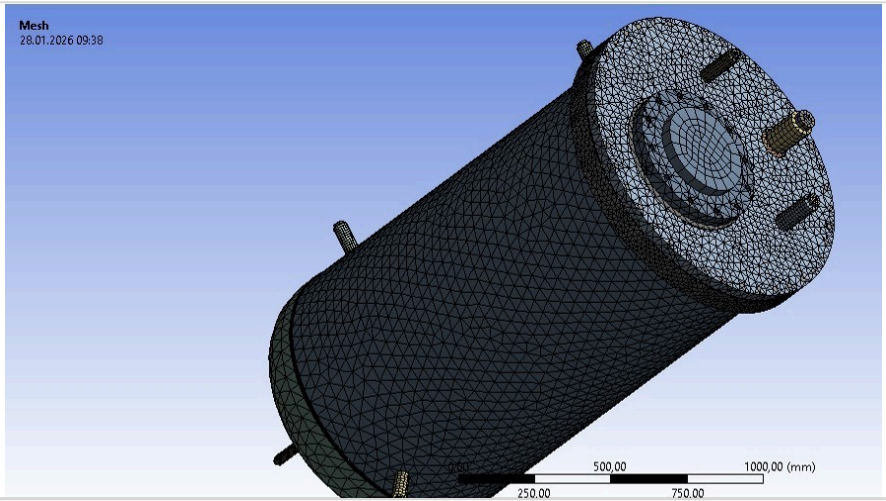
- **Calculated** the shell stresses with thin-wall theory: von Mises 23.3 MPa, radial growth 0.051 mm.
- **Modelled** the vessel and bolted head in ANSYS Mechanical with mesh refinement at the flange and head.
- **Compared** theory with the shell membrane and read the FEA peaks as bending at discontinuities, instead of forcing one number onto both.
- **Extended** the study with stored energy (isentropic gas), TNT equivalent and Hopkinson-Cranz blast distances. Article in preparation.

WHAT I WOULD DO NEXT

A mesh independence study at the peak and a minimum-constraint support instead of the fixed base.



Blast safety distance by pressure scenario (V = 1.1 m³)



Mesh, refined at the discontinuities

A landing gear 50% under its weight target

TEKNOFEST International UAV Competition · Aug 2024 - Aug 2025
Team project · I designed the landing gear

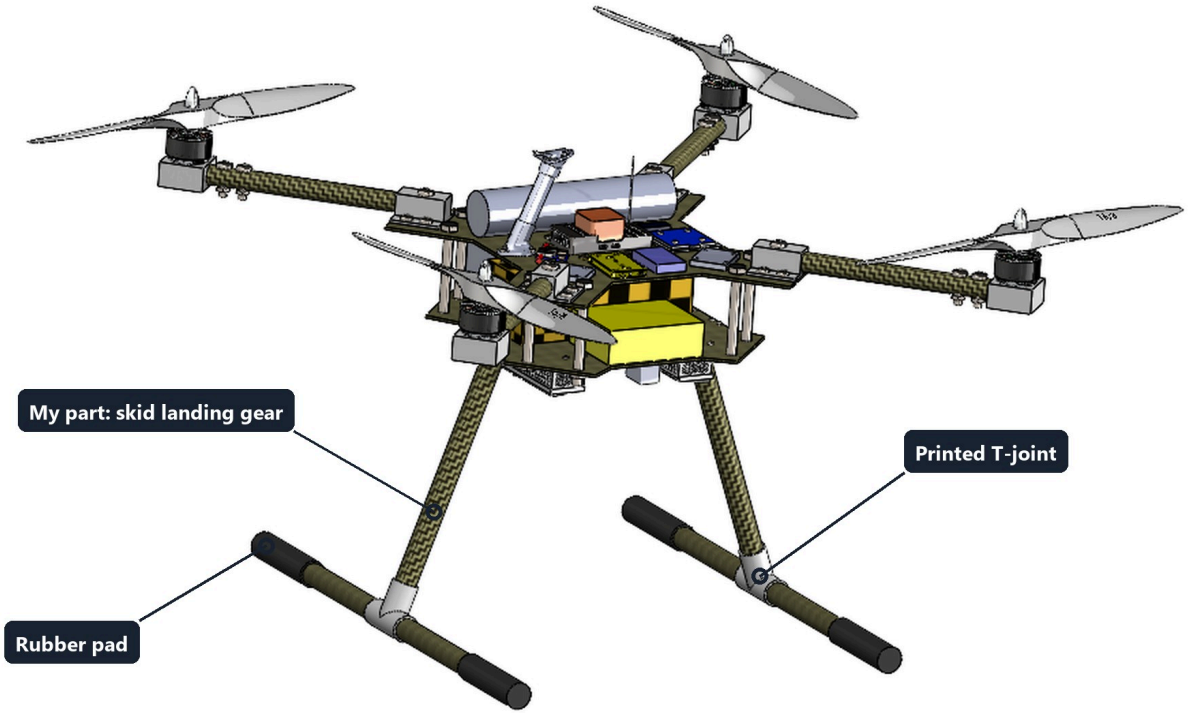
For a 7 kg quadcopter I designed a skid landing gear from carbon-fibre tubes and 3D-printed joints, checked it with static FEA and brought it in 50% below the specification weight target.

50%
below the specification weight target

7.05 kg
take-off mass of the UAV

292 mm
landing gear height

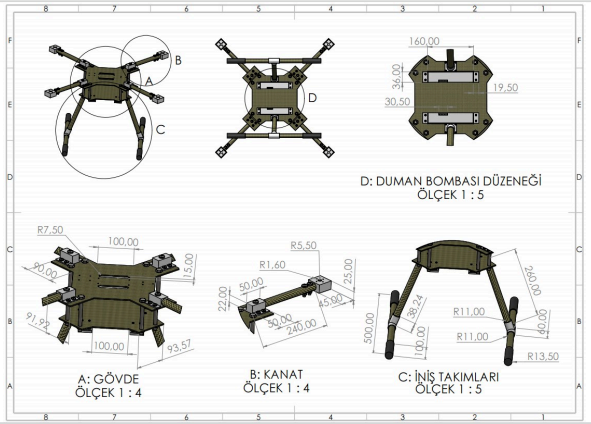
415 mm
landing gear width



The whole system: UAV assembly in SolidWorks



My subsystem: skid legs, carbon tubes, printed T-joints



Drawing sheet with gear dimensions

Light, stiff and easy to fit

1 · REQUIREMENTS
7 kg take-off mass, camera clearance, hard landings

→

2 · PARAMETRIC CAD
Skid legs, T-joints and flanges in SolidWorks

→

3 · STATIC FEA
Landing loads on the gear in ANSYS

→

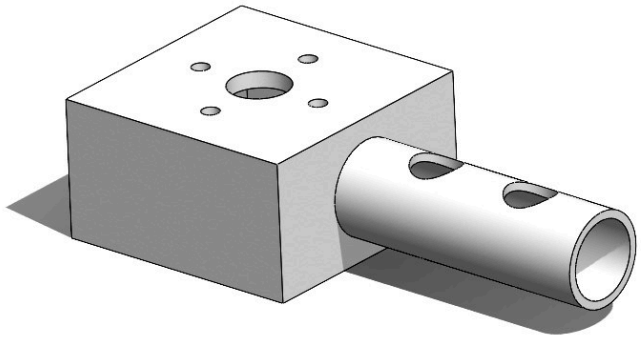
4 · BUILD
Carbon tubes, FDM joints in PLA, ABS and ASA

DESIGN DECISIONS

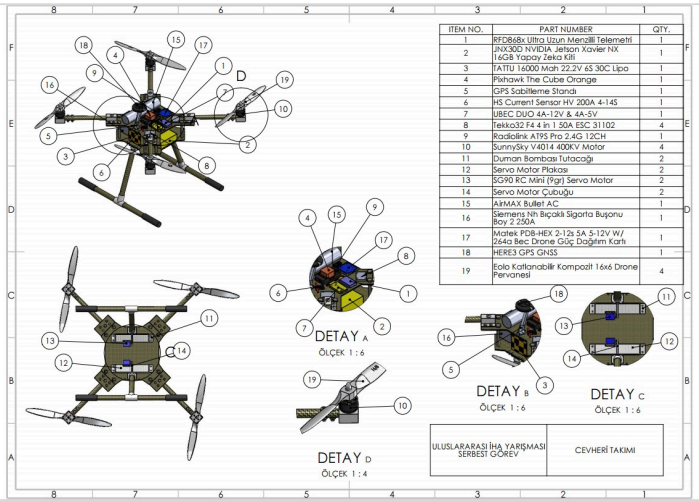
- **Skid gear** instead of arm-mounted legs: out of the camera's view and spreads the landing load over a larger area.
- **Carbon-fibre tubes** with a 5 mm wall on the legs, thicker than the arms, from a local manufacturer.
- **3D-printed T-joints and angled flanges** in ABS for strength-to-weight and easy printing.
- **Rubber pads** on the skids to soften the landing shock.
- **Bolted to the frame** like the other parts, for easy assembly.

SKILLS

Parametric CAD · Static FEA · FDM printing · Material selection · Drawings



3D-printed joint detail



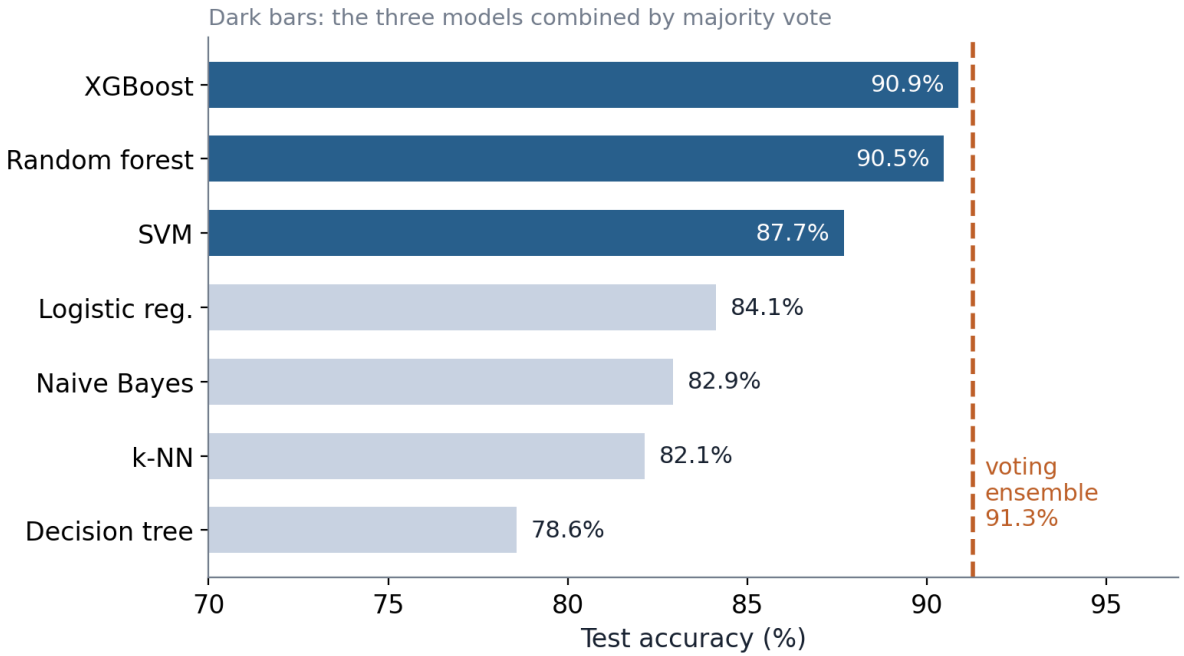
Assembly drawing with bill of materials

Telling steel plate defects apart at 91.3%

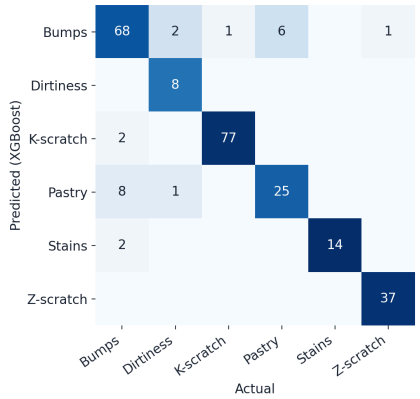
Machine learning course project · Feb 2025 - Jun 2025
Individual work · R

I compared seven classifiers on steel plate inspection data and combined the best three into a voting ensemble that identifies the fault type with 91.3% accuracy.

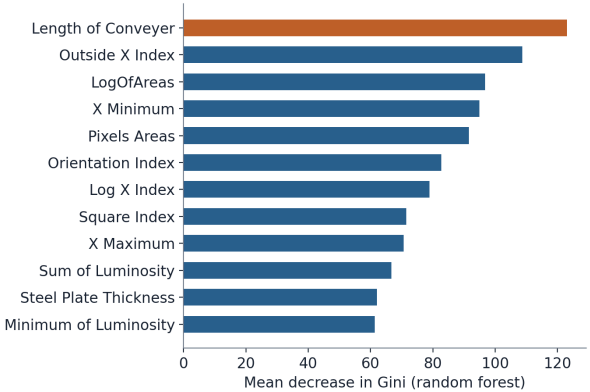
91.3%	0.895	7	27
voting ensemble accuracy	macro F1-score	models compared	input features



Test accuracy of the seven models



Confusion matrix (XGBoost, test set)



Top 12 features (random forest)

From raw measurements to a voting ensemble

1 · DATA
UCI Steel Plates Faults, 27 features per plate

→

2 · PREPARATION
80/20 split, scaling, oversampling, Boruta selection

→

3 · SEVEN MODELS
From logistic regression to XGBoost

→

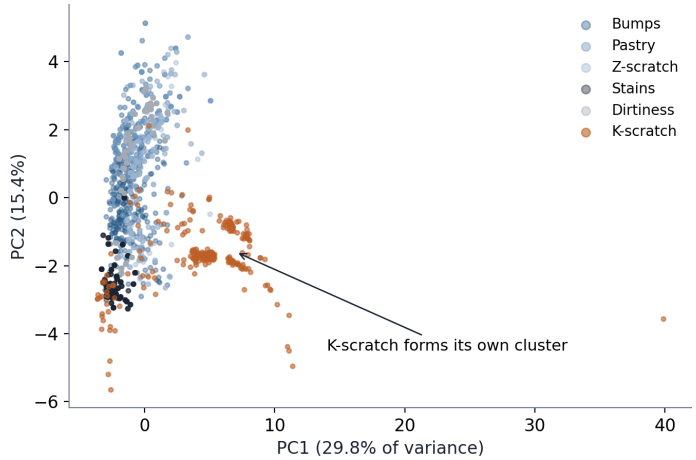
4 · ENSEMBLE
Majority vote of XGBoost, random forest and SVM

WHAT I DID

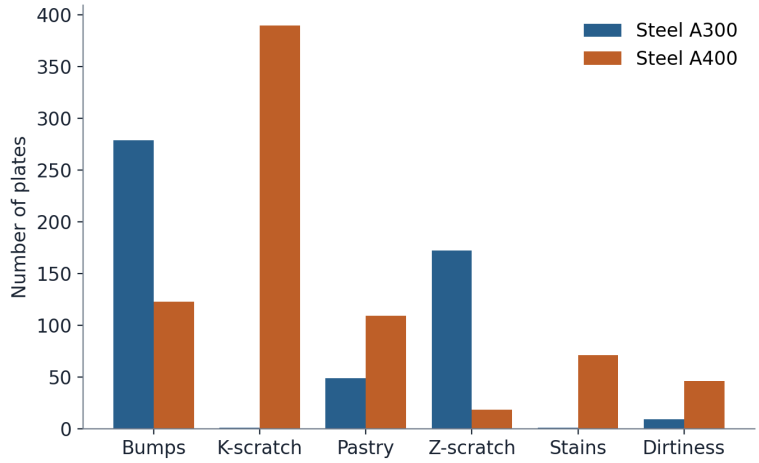
- **Cleaned** the data, left out the ambiguous "other faults" class and balanced the minority classes by oversampling.
- **Selected** the informative features with Boruta.
- **Compared** logistic regression, decision tree, random forest, k-NN, naive Bayes, SVM (5-fold CV for C) and XGBoost.
- **Combined** XGBoost (91%), random forest (90%) and SVM (88%) by majority vote: 91.3%, precision 0.909, recall 0.891.

WHAT IT SAYS ABOUT THE PROCESS

K-scratch, Z-scratch and stains are recognised almost perfectly. The pair the model mixes up most is bumps and pastry. Conveyor length ranks highest, so the model leans on line and plate geometry measurements, and the fault mix differs clearly between the two steel grades.



PCA view of the training data



Fault types by steel grade: K-scratch appears almost only on A400

MORE WORK

Formula Student

İÜC RACING FORMULA STUDENT · POWERTRAIN TEAM · 2023 - 2026

Designed the chain guard in SolidWorks and verified its strength with static analysis in ANSYS. Revised the bracket design for a **15% weight reduction**.

SolidWorks | Static FEA | Lightweighting

INTERNSHIPS

- **FESTO**, 2025: traced mechanical, pneumatic, electrical and software faults in a robotic system and recommissioned it.
- **BEKO**, manufacturing, 2025: found loss sources at 5 critical assembly stations and proposed 2 improvements.
- **Midas Makina**, design, 2024: revised SolidWorks models and prepared manufacturing drawings.
- **Yılmaz Redüktör**, workshop, 2024: machining, heat treatment and gear quality control on site.

SKILLS

SolidWorks (CSWP), ANSYS Mechanical, AutoCAD, Siemens NX, CATIA, Fusion 360, MATLAB/Simulink, R, MS Excel. Static and modal FEA, topology optimisation, GD&T, design for manufacturability, injection moulding and sheet metal design, root cause analysis, lean manufacturing.

CONTACT

mmert1erdogan@gmail.com · linkedin.com/in/mmerterdogan · mmerterdogan.github.io